SRUTHI GADE, AI/ML Engineer
Sruthig.dev26@gmail.com | +1 331 (454) -4770
https://www.linkedin.com/in/gadesruthi/
Open to USA — Onsite / Hybrid (W2 & Full-time Roles)

Professional Summary

Innovative and detail-driven AI/ML Engineer with 3+ years of experience designing and deploying data-driven and Generative AI solutions that deliver measurable business impact. Skilled in developing Large Language Model (LLM) and Retrieval-Augmented Generation (RAG) systems using Python, Lang Chain, and Semantic Kernel, with a strong focus on reliability, scalability, and ethical AI design. Adept at building end-to-end MLOps pipelines and deploying models across AWS, Azure, and GCP environments to support high-performance automation, risk detection, and compliance workflows. Passionate about creating AI solutions that are not only intelligent but also transparent, explainable, and aligned with real-world goals.

Education

- Master of Science in Computer and Information Science, Concordia University - May 2025
- Bachelor of Technology in Computer Science, SR Engineering College - June 2022

Certifications

- Azure AI Fundamentals - Microsoft Certified | 2025
- Generative AI Fundamentals – Databricks | 2025

Technical Skills

- **Programming Languages:** Python, C#, SQL
- **Python Libraries & Tools:** NumPy, pandas, PySpark, scikit-learn, TensorFlow, PyTorch, asyncio, regex, Flask, Fast API, Stream lit
- **Machine Learning Frameworks:** TensorFlow, PyTorch, Scikit-learn, Hugging Face Transformers, XGBoost, LightGBM
- **Generative AI & NLP Tools:** Lang Chain, RAG, Prompt Engineering, AutoGPT, Crew AI, Semantic Kernel, spaCy, NLTK
- **MLOps & Automation:** MLflow, Airflow, Fast API, Flask, Docker, Kubernetes, GitHub Actions, Databricks, Kafka, Azure Data Factory
- **Cloud Platforms**
  - **AWS:** SageMaker, Lambda, Glue, S3, EC2
  - **Azure:** Azure ML, Azure OpenAI, Power Automate, Power Apps
  - **GCP:** Vertex AI, Big Query, Dataflow, Pub/Sub
- **ETL & Data Engineering:** Pandas, NumPy, SQL, PySpark
- **Visualization Tools:** Power BI, Plotly, Matplotlib, Seaborn
- **Databases:** PostgreSQL, MongoDB, Redis, SQL Server
- **Compliance & Governance:** SOX, GDPR, FINRA, Model Risk Management (MRM), Explainable AI (XAI), Data Privacy & Masking

Molina Health care, Remote | AI/ML Engineer **July 2025- Till date**

Overview: Developed AI-driven compliance automation and patient data protection solutions using Generative AI and MLOps practices to enhance regulatory adherence (HIPAA, GDPR) and model transparency across healthcare data pipelines. Built and deployed responsible AI workflows supporting risk assessment, compliance auditing, and explainable clinical AI systems.

Key Responsibilities

- Designed and implemented RAG-based compliance assistants leveraging LangChain, Semantic Kernel, and Azure OpenAI to automate regulatory checks on patient data.
- Built and fine-tuned transformer-based NLP models (BERT, GPT-3.5) for clinical note summarization, entity recognition, and compliance tagging.
- Developed Explainable AI (XAI) dashboards using SHAP and LIME for model interpretability during regulatory audits.
- Created MLOps pipelines with MLflow, Airflow, and GitHub Actions to automate retraining, compliance logging, and version control of healthcare ML models.
- Implemented data anonymization and synthetic data generation using PyTorch and scikit-learn for HIPAA-compliant AI model training.
- Integrated Azure Data Factory and Databricks for real-time ingestion of clinical data, ensuring traceability and governance.
- Collaborated with compliance and data privacy teams to align ML workflows with SOX, HIPAA, and GDPR regulations.
- Applied model risk management (MRM) and bias detection to ensure fairness and transparency in patient data processing.

Key Achievements

- Automated over 60% of manual compliance documentation using AI copilots, reducing review times by 35%.
- Improved audit traceability and explainability across ML models, meeting HIPAA and MRM standards.
- Deployed compliant AI pipelines that enhanced healthcare data governance and model accountability.

Bank of America, IL | AI/ML Engineer **Mar 2024 – May 2025**

Overview: Currently developing and deploying AI-driven fraud detection and compliance automation pipelines leveraging Generative AI, LLMs, Python, and MLOps practices across AWS and Azure cloud environments.Built ML models for fraud detection, AML (Anti-Money Laundering), and credit scoring, aligned with SOX and FINRA compliance.

Key Responsibilities

- Designing RAG-based AI pipelines using LangChain, Python, and FAISS for real-time fraud detection and compliance automation.
- Fine-tuning transformer-based LLMs (BERT, RoBERTa, Falcon) using Hugging Face and PyTorch for entity extraction and document analysis.

- Developed AI copilots using Semantic Kernel to orchestrate multi-step workflows with Azure OpenAI, improving task automation efficiency by 35%.
- Integrated RAG workflow with enterprise knowledge bases (SharePoint, Confluence, internal APIs) to automate knowledge retrieval and reduce manual effort by 40%.
- Deployed LLM and ML models using FastAPI microservices with Docker, Kubernetes, reducing deployment time by 50%.
- Implemented model monitoring & drift detection using MLflow, Prometheus, SHAP to ensure long-term model performance and compliance.
- Designed and deployed LLM-powered applications using OpenAI GPT, Azure OpenAI, and Llama 2/3 for enterprise use cases.
- Secured GCP services using IAM roles, service accounts, VPC Service Controls, and Secret Manager.
- Integrated Semantic Kernel plugins with enterprise microservices for fraud investigation, compliance analysis, and risk scoring.
- Writing efficient Python scripts for data preprocessing, model evaluation, and API integration.
- Implementing data ingestion and transformation pipelines using AWS Glue, Azure Data Factory, and pandas.
- Developing FastAPI-based microservices in Python for low-latency model inference integrated into enterprise systems.
- Automating MLOps pipelines with Python, MLflow, Airflow, and GitHub Actions for retraining, monitoring, and deployment.
- Applying Explainable AI (SHAP, LIME) for transparent model interpretation in compliance reviews.
- Developed E2E RAG evaluation pipeline including retrieval precision, context relevancy, and answer faithfulness scoring.
- Using Python asynchronous programming with asyncio for large-scale text and document processing.
- Built Semantic Function pipelines to combine prompt templates, embeddings, and planner logic for reliable task decomposition.
- Building multi-cloud AI workflows across AWS, Azure ML, and GCP Vertex AI for scalable deployments.
- Leveraging Databricks, Kafka, and PySpark for distributed data processing and real-time event analytics.
- Conducting bias detection, drift tracking, and retraining in compliance with MRM standards.
- Collaborating with security teams to ensure data encryption, IAM enforcement, and secure AI deployment under GDPR and SOX.

Key Achievements

- Prevented over $3M in fraudulent transactions annually using AI-driven anomaly detection.
- Reduced compliance audit time by 40% via automated explainability and reporting scripts.
- Decreased retraining cycle duration by 25% using optimized Python pipelines and CI/CD integration.

TetraSoft, India | Machine Learning Engineer **Feb 2022 – May 2023**

Overview: Developed and deployed predictive analytics and data warehousing solutions for retail clients to forecast demand, improve sales planning, and strengthen data governance practices.

Key Responsibilities

- Built data warehousing pipelines for retail data integration using SQL, Azure Synapse, and Snowflake.
- Designed ETL workflows using pandas, PySpark, and Airflow for structured and unstructured retail datasets.
- Developed predictive ML models using TensorFlow, Scikit-learn, and XGBoost for forecasting and churn prediction.
- Established data governance standards ensuring data lineage, accuracy, and consistency across ML workflows.
- Deployed Dockerized APIs using Flask and Kubernetes for scalable ML services.
- Applied data validation checks using Great Expectations to ensure data quality in ML pipelines before ingestion.
- Automated retraining and deployment cycles using MLflow, GitHub Actions, and Airflow DAGs.
- Created Power BI dashboards to visualize KPIs, sales trends, and model performance.
- Implemented data anonymization, metadata management, and quality validation aligned with GDPR and enterprise data governance frameworks.
- Collaborated with business and data engineering teams to maintain data cataloging and retention policies for compliance.
- Optimized model performance using feature engineering, cross-validation, and hyperparameter tuning.

Key Achievements

- Designed a unified data warehouse architecture that improved data accessibility and accuracy by 30%.
- Strengthened data quality and consistency through well-defined data governance frameworks.
- Improved forecasting precision and operational planning across the retail domain.

Project Highlights

Responsible Compliance AI Platform – Molina Healthcare

Engineered an AI-driven healthcare compliance solution using RAG, Azure OpenAI, and secure MLOps to automate HIPAA/GDPR checks and clinical document validation with NLP.

Impact: Reduced manual compliance review efforts by 60% and accelerated audit processes by 35% with explainable, fair, and traceable healthcare AI workflows.

AI-Powered Fraud Detection Platform – Bank of America

Developing a Python-based Generative AI fraud detection solution integrating LLMs, RAG, and MLOps for compliance monitoring and risk scoring.

Impact: Prevented $3M+ in fraud annually and improved compliance transparency by 40%.

Retail Forecasting System – TetraSoft

Built Python-driven predictive analytics models using Scikit-learn, TensorFlow, and XGBoost to forecast demand and optimize stock planning.

Impact: Increased forecast accuracy by 20% and improved retail decision-making speed.